JAMES PAOLO RILI

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SKILLS

Programming Languages	C, Rust, OCaml, MATLAB, Python, Java, Kotlin, TypeScript,
Libraries	NumPy, Matplotlib, Pandas, SciPy, scikit-learn, Keras
Frameworks	Compose, Express.js, React.js, Qt
Database	PostgreSQL, MongoDB, MySQL
Hardware	Electrical & RF Equipment, NI-VISA, Arduino, Raspberry Pi
Other Tools	Linux, Vim, AWS, Firebase, Git/GitHub, Atlassian, Microsoft Office

WORK EXPERIENCE

Software Developer Co-op, Reporting Services | Caracal Technologies | *Vancouver, BC* — May 2025 - Dec 2025

- Engineered new features for a web application processing high-volume transit data, delivering automated and accurate financial reporting for enterprise clients using Python via AWS Lambda for the backend and TypeScript with React.js on the frontend.
- Refactored legacy modules and queries by applying functional and object-oriented principles, leading to increased system scalability and future maintainability.
- Streamlined workflow in Bitbucket by writing Bash scripts and documenting standard procedures, reducing the risk of human error during deployment.

RF Software Engineer Co-op | NETGEAR, Inc. | *Richmond, BC* — Sep 2022 - Apr 2023

- Developed and optimised Python-based software for WWAN validation of hotspot devices, reducing validation time by 50%.
- Spearheaded the implementation of coding standards and led a comprehensive refactoring initiative, enhancing software scalability and maintainability.
- Wrote documentation for users and developers of the software environment, reducing onboarding time from 6 to 3 days.

Physics Teaching Assistant | University of British Columbia | *Vancouver, BC* — Jan - Apr 2024

- Conducted weekly workshops for 50 undergraduate students to reinforce understanding of complex problems.
- Delivered concise lectures about waves and basic hydrostatics to ensure material comprehension.
- Provided one-on-one assistance during office hours, assessing individual learning needs and reinforcing key concepts.

EDUCATION

Computer Systems Technology Diploma — Sep 2024 - Present
British Columbia Institute of Technology — CGPA: 93%

- Functional Programming
- Databases
- Operating Systems
- System Design
- OOP
- Business Communications

Bachelor of Science in Physics — Sep 2019 - Apr 2024
University of British Columbia

- Experimental Physics
- Data Science
- Statistical Mechanics
- Computational Physics
- Software Construction
- Writing for Research

PROJECTS

Iron-OBD2 | BCIT + Iron IoT (https://github.com/ironiot/iron-obd2/) — Apr - May 2026

- Led a team of five in order to build a MVP of cloud-based OBD-II software, adding extended capabilities for vehicle diagnostics.
- Communicated with client and teammates to ensure specifications are met and project on track to finish in under one month.
- Wrote unit-testing with Python and automation code with Bash, accelerating deployment while maintaining bug-free software.

Student Management Information System | BCIT Jan - Feb 2025

- Built a custom database engine in C by implementing CRUD to handle student records via file I/O and memory management.
- Designed a command-line interface with the REPL pattern, leading to a modular codebase.
- Implemented error handling and input validation to ensure data integrity and system reliability.

EcoMeter | Enactus BCIT (https://www.eco-meter.ca/) Sep 2024 - Apr 2025

- Assist the development team to maintain and optimise a student-run sustainable dining guide with Ruby on Rails.
- Integrate Mixpanel to provide user behaviour and engagement metrics for the club's data and marketing teams.
- Research and evaluate available technologies to identify opportunities for website performance and functionality improvements.

Machine Learning Interatomic Potentials | UBC (https://arxiv.org/abs/2404.18393) Apr 2024

- Engineered a neural-network model with Keras API and object-oriented design to accurately predict crystalline atomic energies, providing a computationally efficient alternative to conventional, physics-based methods.
- Accelerated the search for the optimal model with feature engineering, cross-validation, and hyperparameter tuning.

Classification of Habitable Planets | UBC Feb 2024

- Applied different machine learning classification algorithms to the NASA exoplanet dataset to determine most accurate model for determining habitable planets.
- Automated the model selection with Python through scikit-learn, along with NumPy and Pandas through Jupyter.

Characterisation of Semiconducting Materials | UBC (https://arxiv.org/abs/2312.05744) Sep - Dec 2023

- Extracted material information of a semiconducting sheet with a specific four-point probe technique.
- Compiled a scientific paper on the results and its implications for future quantum computing hardware.
- Employed the use of NumPy and Pandas for data processing, analysis, and characterisation of semiconductor properties.

Numerical Simulations of Physical Phenomena | UBC Jun 2023

- Wrote a program with MATLAB that simulates the physics of two colliding galaxies.
- Developed a physical model in Python that animates the behaviour of electron spin in a magnetic sheet.
- Designed and wrote simulations of 2-dimensional quantum particles in MATLAB to observe tunneling behaviour.

Circuit Prototyping Mini-Projects | Personal Aug 2022

- Prototyped circuits with a breadboard, micro-controller, and sensing components to obtain data for physics experiments.
- Employed statistical techniques to reduce measurement uncertainty by 35%.
- Programmed the behaviour of components with the Arduino IDE and processed data with Python for fast data analysis.

Digit Recogniser | Personal Jul 2022

- Built, trained, and tested a neural network that recognises handwritten digits in Python.
- Written with only the use of NumPy and linear algebra to understand fundamental concepts of neural networks.

Analysis of Johnson-Nyquist Noise in Electrical Components | UBC Sep - Dec 2021

- Performed a series of experiments in an electrical laboratory investigating the effect of cosmic background noise on resistors.
- Analysed data using NumPy and Pandas, and configured calculations for theoretical models with SciPy.
- Composed a research paper detailing the process and presented the results to an audience of researchers and physicists.

ORGANISATIONS

UBC Physics Society Sep 2019 - Apr 2024
General Member

Int'l Association of Physics Students Sep 2022 - Apr 2024
General Member